Sho Sakaue

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Education

University of Utah — Salt Lake City, Utah
Bachelor of Science in Computer Science and Applied Mathematics — *Aug. 2024 - Expected Dec. 2027*

- **GPA**: 3.96/4.0
- **Relevant Coursework**:
  - **CS**: Machine Learning, Deep Learning, Probabilistic Deep Learning, Natural Language Processing, Computer Vision, Computational Geometry, Data Structures & Algorithms, Models of Computation
  - **Math**: Optimization, Numerical Analysis, Probability Theory & Statistics, Linear Algebra, Real Analysis, Calculus I–III
  - **Large Language Models**: - Matsuo Lab, University of Tokyo — June. 2026 - Present

Research Experience

Undergraduate Research Assistant - *under Professor Suhas Prameela* — Aug. 2025 - Present
University of Utah — *Salt Lake City, Utah*

- Generated training dataset for Machine Learning Interatomic Potentials(MLIPs) across defects, surfaces, and alloy configurations from DFT calculations.
- Trained MLIPs on DFT reference datasets, and increased convergence speed by day selecting better optimization method.
- increased model prediction accuracy by 15 % on avergae while maintaining inference speed

Conference & Presentation

TMS 2027 Annual Meeting & Exhibition - *Accepted for Oral Presentation* — Mar. 2027

- Abstract accepted for oral presentation: *Particle shape and interface chemistry control defect relaxation in GRX-810 ODS alloys.*
- Co-authored research examining how particle morphology and oxide–matrix interface chemistry influence defect relaxation in GRX-810 oxide dispersion strengthened alloys.

Undergraduate Research Symposium - University of Utah — July. 2026

- Presented my work supported by Undergraduate Research Oppotunity Program (UROP) funding
- *Machine Learning Interatomic Potentials for NiCoCr-based alloys*

Work Experience

Computer Science Teaching Assistant - CS 1400 (Intro to Computer Programming) — Jul. 2026 - Present
University of Utah — *Salt Lake City, Utah*

- Held weekly group lab session with 30+ students to help them master coding fundamentals and cover where lecture does not
- Graded 100+ students submissions with detailed feedback to increase students performance in the class

Mathematics Tutor and Grader — Aug. 2025 - July. 2026
University of Utah — *Salt Lake City, Utah*

- Worked with 10+ professors to grade assignments and exams across 1,000+ student submissions, providing detailed feedback to enhance academic performance.
- Tutored students in Calculus I, II, III, Linear Algebra, and Statistics through clear explanations for each student.
- Adapted strategies to each student's learning style to improve understanding and problem-solving independence.

Projects

BayesBench | *numpy, scikit-learn, matplotlib, pandas* June. 2026 - Present

- Implemented the benchmark of Bayesian Optimization method for various limited-budget black-box optimization.
- Added experiment scripts and plotting tools to compare optimization performance across methods.
- Structured the project as a reusable Python package for future extension to more benchmarks and optimizers.

LoRA and QLoRA for Fine-Tuning (Paper reproduction) | *PyTorch, Hugging Face* Mar. 2026 – Apr. 2026

- Reproduced the LoRA and QLoRA methods from their original research papers, translating the proposed low-rank adaptation and quantized fine-tuning techniques into a working training pipeline.
- Fine-tuned a llama-2-1B and benchmarked LoRA and QLoRA against the base model and full fine-tuning, comparing accuracy-performance tradeoffs.
- Conducted hyperparameter experiments across LoRA ranks and quantization bit widths to analyze their effects.

Multiplayer Networked Game & HTTP Server - *Software Practice* | *C#, SQL* Nov. 2025 - Dec. 2025

- Developed a distributed multiplayer game client in Blazor using the MVC architectural pattern.
- Implemented a multi-threaded TCP network controller to communicate by JSON, using locks to ensure thread safety and prevent race conditions.
- Integrated a MySQL database to store game history and player high scores.

Technical Skills

Programming Languages : Python, C/C++, Java, Bash, SQL(MySQL), C#
Markup & Style Languages: HTML, CSS, LaTeX, Markdown
Machine Learning: PyTorch, NumPy, Hugging Face, scikit-learn, SciPy, Matplotlib, Pandas
Developer Tools: Git(GitHub, GitLab), Docker, Linux, HPC/SLURM computing

Competition/Hackathon

MIT iQuHACK (Quantum Computer) Feb. 2026

- Simulated 50–60 qubit quantum circuits using Qiskit to identify hidden peak bitstrings, placing 24th in BlueQubit Challenge.
- Optimized quantum circuits simulation performance by gate reordering, reducing runtime by approximately 30%.
- Developed statistical candidate-selection heuristics to infer peak from near-uniform output distributions.

Crimson Hack (University of Utah) Oct. 2025

- Built a rocket simulation game, where players navigate from one planet to another while avoiding obstacles and debris.
- Implemented planet-specific gravity and fuel-limited movement mechanics considering physical reality.
- Designed interactive gameplay logic that allowed users to select different solar-system planets and control rocket

Awards & Certificates

Frank Stenger Scholarship	Fall 2026
Undergraduate Research Opportunity Program (UROP)	Summer 2026
24 th place in BlueQubit challenge in MIT iQuHack	Feb. 2026
Dean's List	Fall 2024 - Summer 2026
ISSS Scholarship	Spring 2026
Utah Global Scholarship	Fall 2024 - Fall 2027

Language

English - Fluent
Japanese - Native